

Enumerative combinatorics

For a finite set A , we denote:

1) $|A|$ - the number of elements (the cardinality of A)

2) $A \times B = \{(a, b) : a \in A, b \in B\}$ - Cartesian product

3) $A \cup B$ - union

4) $A \cap B$ - intersection

5) $A \setminus B$ - difference

1. Counting by bijections

Def. 1) Let $f: A \rightarrow B$ be a function:

- f is injective if different elements have different images: $x \neq y \Rightarrow f(x) \neq f(y)$

- f is surjective if every element of B is the image of an element of A .

- f is bijective if it is injective and surjective, that is, if every element of B is the image of exactly one element of A .

Obs. A function $f: A \rightarrow B$ is bijective $\Leftrightarrow f$ is invertible, that is, there is $g: B \rightarrow A$ with $g(f(x)) = x, \forall x \in A$ and $f(g(y)) = y, \forall y \in B$.

Def. 2) A set A has n elements, $|A|=n$, if there is a bijection $f: \{1, 2, \dots, n\} \rightarrow A$. The empty set has 0 elements.

Th. 3) (The bijection principle) Let A and B be two finite sets:

a) $|A|=|B| \Leftrightarrow$ there is a bijection $f: A \rightarrow B$

b) $|A| \leq |B| \Leftrightarrow$ there is an injective function

$f: A \rightarrow B$

Method. When a set is hard to count directly, we look for a bijection between it and a set that is easy to count. To show that a correspondence is bijective, we also describe the inverse procedure ("the way back").

ex.: $3, 10, 17, 24, \dots, 2026$ #

$7, 14, 21, 28, \dots, 2030$ /

$1, 2, 3, 4, \dots, 290 \Rightarrow 290$ nr

$$100 \cdot 7 - 1 = 700 - 1 = 696$$

$$f(a) = \frac{(a+1)}{7} \quad f: \{3, 10, \dots, 2026\} \rightarrow \{1, 2, \dots, 290\}$$

Obs. (Watch out for +1). An arithmetic progression $a, a+n, a+2n, \dots, b$ has $\frac{b-a}{n} + 1$ terms.

$$\begin{aligned} \text{ex.: } 3^0 &= 1 \\ 3^1 &= 3 \\ 3^0 + 3^1 &= 4 \end{aligned}$$

$$\begin{aligned} 2^0 &= 1 \\ 2^1 &= 2 \\ 2^0 + 2^1 &= 3 \end{aligned}$$

$$\begin{aligned} 3^2 &= 9 \\ 3^0 + 3^2 &= 10 \\ 3^1 + 3^2 &= 12 \\ 3^0 + 3^1 + 3^2 &= 13 \end{aligned}$$

$$\begin{aligned} 2^2 &= 4 \\ 2^0 + 2^2 &= 5 \\ 2^1 + 2^2 &= 6 \\ 2^0 + 2^1 + 2^2 &= 7 \end{aligned}$$

$$202600_{(10)} = 161021220201_{(3)}$$

202600	3
67533	1
22511	0
7503	2
2501	0
833	2
277	2
92	1
36	2
10	0
3	1
1	0
0	1

$$\Rightarrow 101011111111_{(3)} =$$

$$= 200110_{(10)}$$

$$= 2^{11} + 2^9 + 2^7 + 2^6 + \dots + 2^0$$

$$= 2815 \text{ number}$$

$$2026_{(10)} = 1111101010_{(2)} = 2812_{(10)}$$

2. The product rule (the rule of "and")

Ex. $A \rightarrow B$ 6 routes $\Rightarrow 6 \cdot 5 = 30$ possibilities
 $B \rightarrow C$ 5 routes

Th. 1) (The product rule) For any finite sets $X_1,$

$X_2, \dots, X_n$:

$$|X_1 \times X_2 \times \dots \times X_n| = |X_1| \cdot |X_2| \cdot \dots \cdot |X_n|$$

2) (The product rule, in words) Each object of a set M is built by a sequence of n choices, made one after the other, so that:

(i) at step i there are exactly m_i options, whatever the choices at the previous step.

(ii) each object of M is obtained from exactly one sequence of choices.

$$\text{Then } |M| = m_1 \cdot m_2 \cdot \dots \cdot m_n$$

$$\text{ex.: } \underline{9} \underline{9} \underline{8} = 9 \cdot 9 \cdot 8 = 648 \text{ de numere}$$

3. Double counting and the division rule:

ex. A, B, C, D, E, F, G, H

$$\begin{array}{l} 1 \text{ player} \rightarrow 7 \text{ pos.} \\ 2 \text{ player} \rightarrow 5 \text{ pos.} \\ 3 \text{ player} \rightarrow 3 \text{ pos.} \\ 1 \text{ player} \rightarrow 1 \text{ pos.} \end{array} \quad \left| \Rightarrow 7 \cdot 5 \cdot 3 \cdot 1 = 105 \right.$$

A, B, C, D

$$\begin{array}{l} 1 \text{ player} \rightarrow 3 \text{ pos.} \\ 2 \text{ player} \rightarrow 1 \text{ pos.} \end{array} \quad \left| \right.$$

$$\Rightarrow 3 \cdot 1 = 3$$

Ob. Where there is no order, we impose one.

Th. 1) (The division rule) If $f: A \rightarrow B$ and every element of B is the image of exactly $k \geq 1$ elements of A , then $|A| = k|B| \Rightarrow |B| = \frac{|A|}{k}$.

Obs. 1) A set with n elements has 2^n subsets

2) $2m$ people can be split into pairs in:

$$\frac{(2m)!}{2^m \cdot m!} = (2m-1)!! = (2m-1)(2m-3)\dots 3 \cdot 1 \text{ ways}$$